Falcon 9 First-Stage Landing Prediction

Data Science Capstone Project Report

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Historical launch analysis, interactive analytics and classification

https://github.com/shyrradev/PYTHON/tree/ibm-spacex-capstone/ibm-spacex-capstone

Executive Summary

The main dataset contains 90 Falcon 9 launches from 2010 to 2020. Successful landing outcomes account for 60 launches (66.7%).

Landing outcomes vary by launch site, orbit and mission era. KSC LC 39A records 17 successful outcomes from 22 launches (77.3%).

The decision tree has the strongest training cross-validation score (87.8%). Its held-out accuracy is 77.8% on 18 launches.

The evidence supports historical screening. A small sample and changing mission conditions limit deployment confidence.

IBM course datasets. Calculations reproduced for this report.

Introduction

Business question: can launch characteristics help predict whether a Falcon 9 first stage achieves a successful landing outcome?

A successful recovery can support booster reuse. This analysis estimates landing outcomes rather than launch prices or actual reuse savings.

The workflow combines course data collection snapshots, cleaning, SQL exploration, visual analytics and supervised classification.

Target: Class = 1 for a successful recorded landing outcome. Class = 0 includes failures, no landing attempt and unknown outcomes.

IBM course datasets. Calculations reproduced for this report.

Data Collection – SpaceX API

API workflow: GET /v4/launches/past retrieves launch records. Resolve /rockets/{id}, /payloads/{id}, /launchpads/{id} and /cores/{id} to enrich the records.

Select Falcon 9 launches with a single core. Extract payload mass, orbit, launch site, core reuse information and landing outcomes.

The quantitative analysis uses IBM's published API-derived historical snapshot: 90 Falcon 9 launches, 4 June 2010 to 5 November 2020.

The collection notebook documents source URLs and checks the snapshot structure. The live SpaceX API request timed out during reproduction.

GitHub notebook: 01_data_collection.ipynb

IBM course datasets. Calculations reproduced for this report.

Data Collection – Web Scraping

Retrieve the public Falcon 9 / Falcon Heavy launch history page with urllib.request. Save its HTML for parsing.

Parse HTML tables with pandas.read_html and the lxml parser. The retrieved page contains 19 tables.

Inspect flight number, date, booster, launch site, payload, orbit and landing columns. Export the largest launch table as a CSV.

Keep the newly scraped history separate from the 90-launch course snapshot because their time periods and row definitions differ.

GitHub notebook: 02_web_scraping.ipynb

IBM course datasets. Calculations reproduced for this report.

Data Wrangling Methodology

Inspect columns, missing values and categories. Convert launch dates, retain Falcon 9 records and inspect payload masses.

The course snapshot already contains filled payload masses. LandingPad has missing values, which the supplied encoded matrix represents without a category.

Create the binary target from Outcome. True ASDS, True RTLS and True Ocean map to 1. Other recorded outcomes map to 0.

Use the 83-column course feature matrix. Exclude Outcome and Class from predictors. Fit StandardScaler inside each training pipeline.

GitHub notebook: 03_data_wrangling.ipynb

IBM course datasets. Calculations reproduced for this report.

EDA with Data Visualization – Methodology

Pandas groupby computes counts and landing rates by launch site, year, orbit and payload band. Charts retain the denominators.

SQLite loads the separate 101-row SQL dataset. Queries use DISTINCT, LIKE, aggregates, subqueries and date filters.

Folium maps launch coordinates and clusters success/failure markers. Tooltips show the site and number of successful outcomes.

Plotly Dash links the site dropdown and payload range to a success-share pie chart and a payload/outcome scatter plot.

GitHub notebook: 04_eda_visualization.ipynb

IBM course datasets. Calculations reproduced for this report.

EDA with SQL – Methodology

Load Spacex.csv into an in-memory SQLite database as launches. Trim whitespace in outcome labels before grouping.

Inspect unique launch-site labels and CCA-prefixed records with DISTINCT and LIKE. Use SUM, AVG and MAX to summarize payloads.

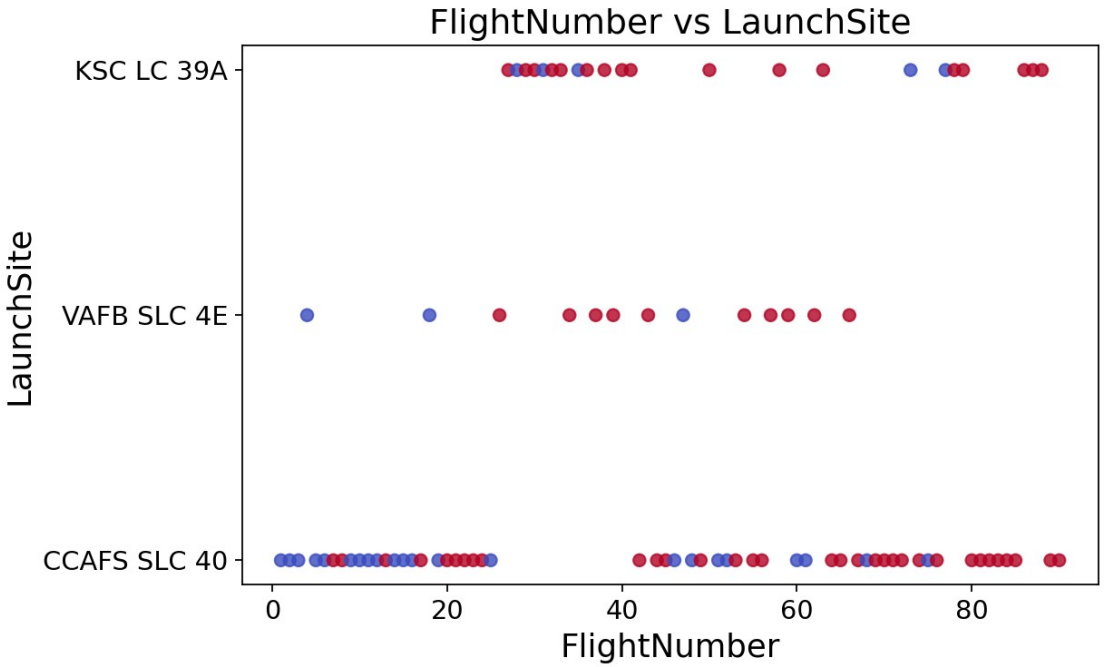
Use MIN(Date) for the first successful ground landing. Filter successful drone landings with payloads between 4,000 and 6,000 kg.

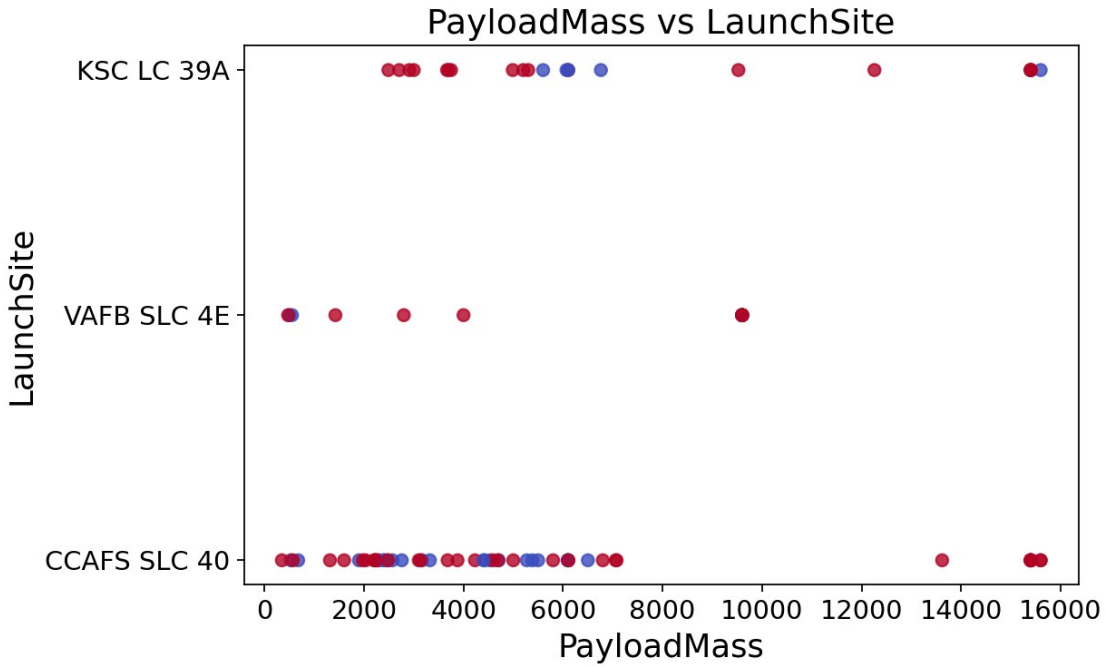
Group mission outcomes, inspect the 2015 drone failures and rank landing outcomes between 2010-06-04 and 2017-03-20.

GitHub notebook: 05_eda_sql.ipynb

IBM course datasets. Calculations reproduced for this report.

EDA Results: Flight Number and Payload by Site



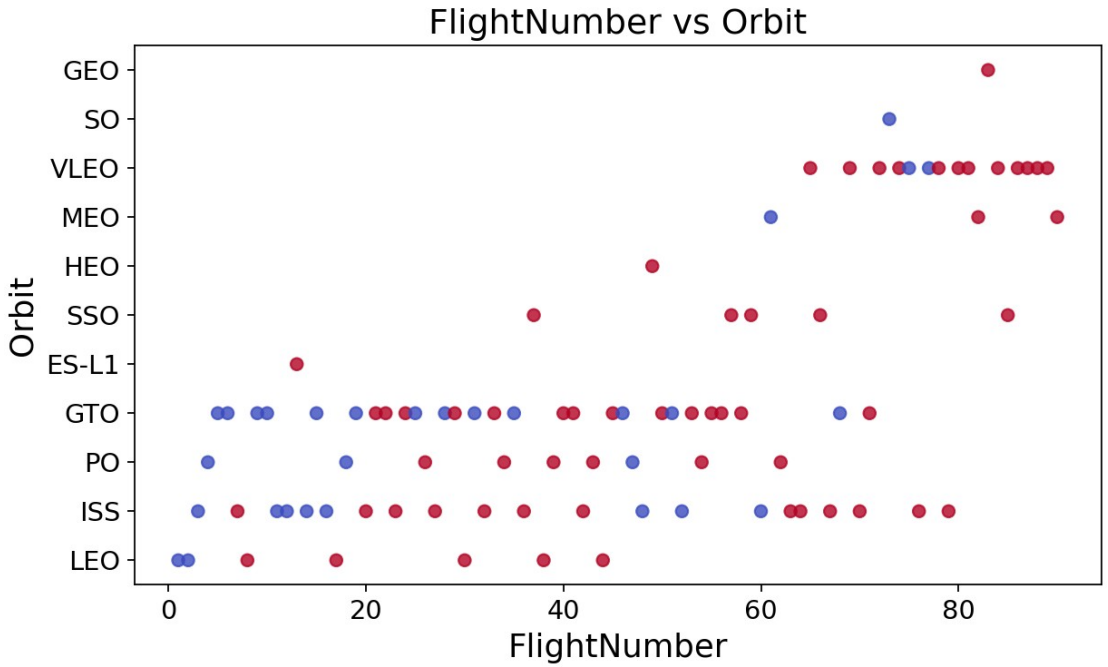


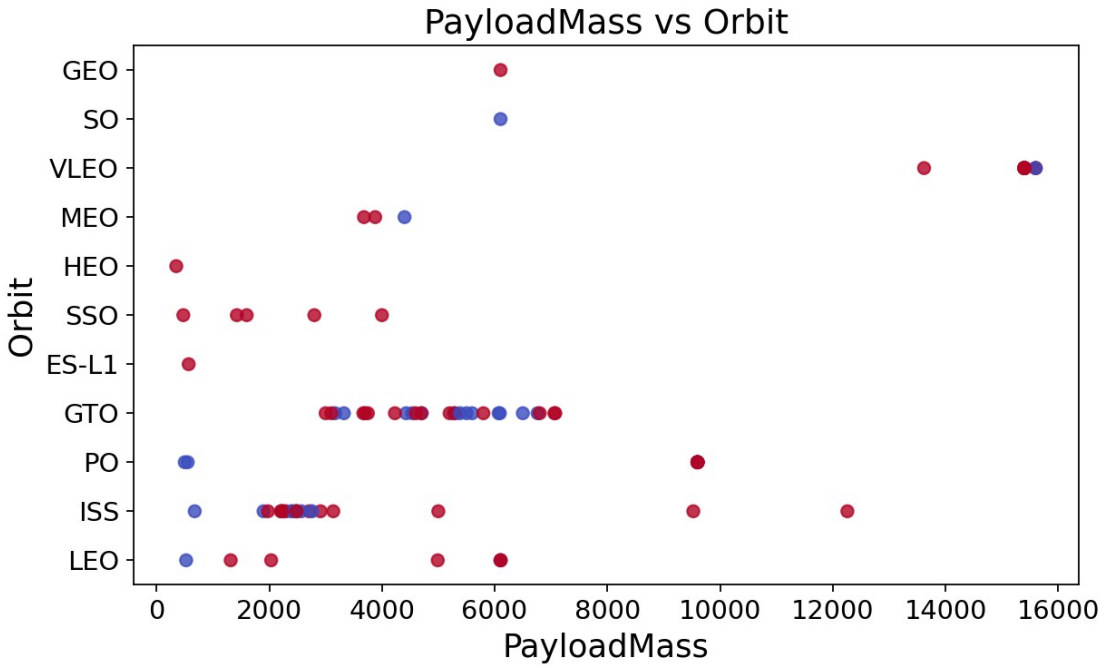
Blue points = class 0. Red points = class 1. Later flight numbers show more successful outcomes. Payload distributions differ across sites, so the site comparison also reflects mission mix.

GitHub: EDA charts and outputs

IBM course datasets. Calculations reproduced for this report.

EDA Results: Flight Number and Payload by Orbit



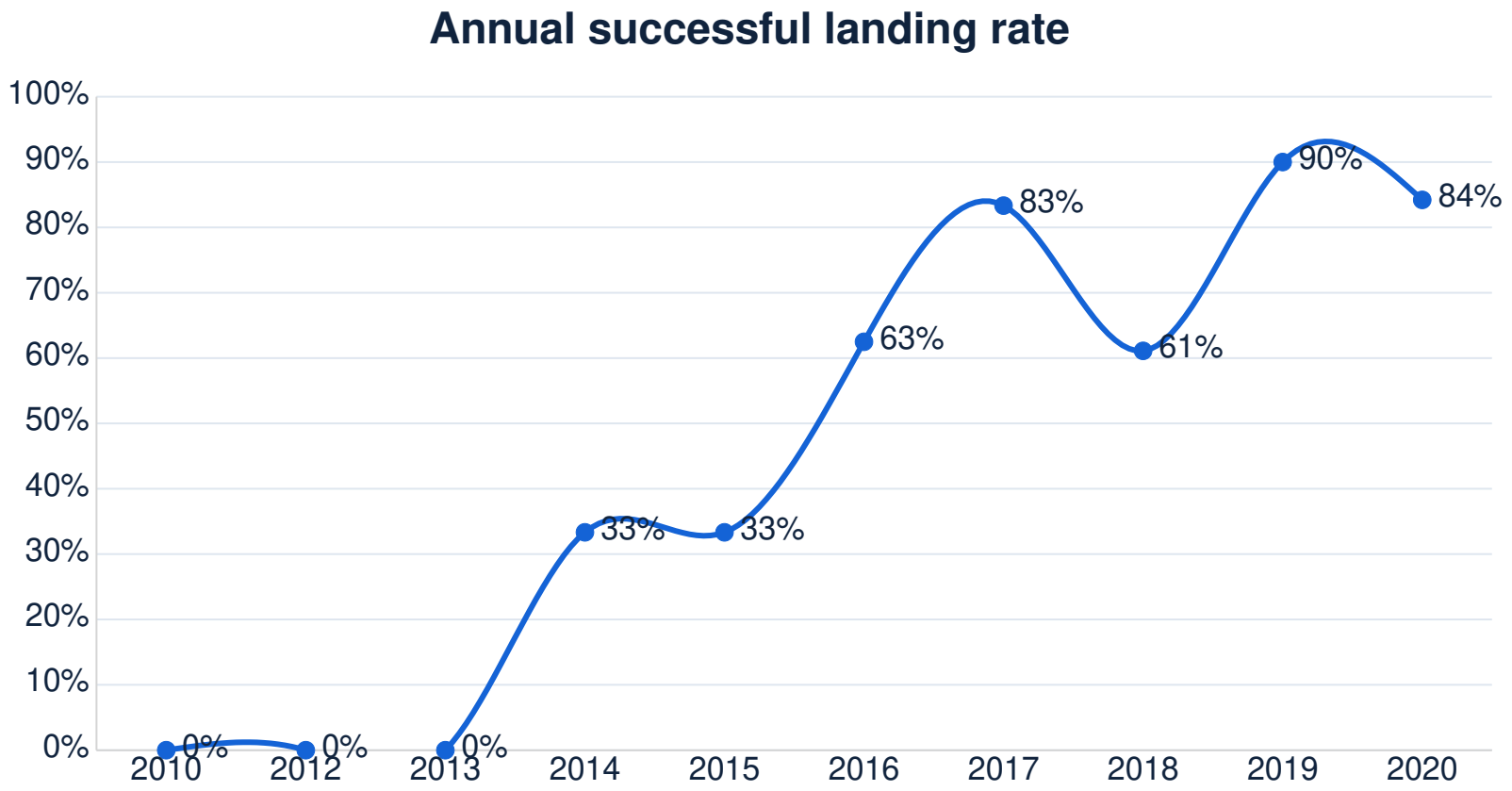


Blue points = class 0. Red points = class 1. Mission orbit and payload are linked. Several small orbit groups contain too few launches for reliable rate comparisons.

GitHub: EDA charts and outputs

IBM course datasets. Calculations reproduced for this report.

Landing Performance Across Years



2010–2013
0 successful / 5 launches

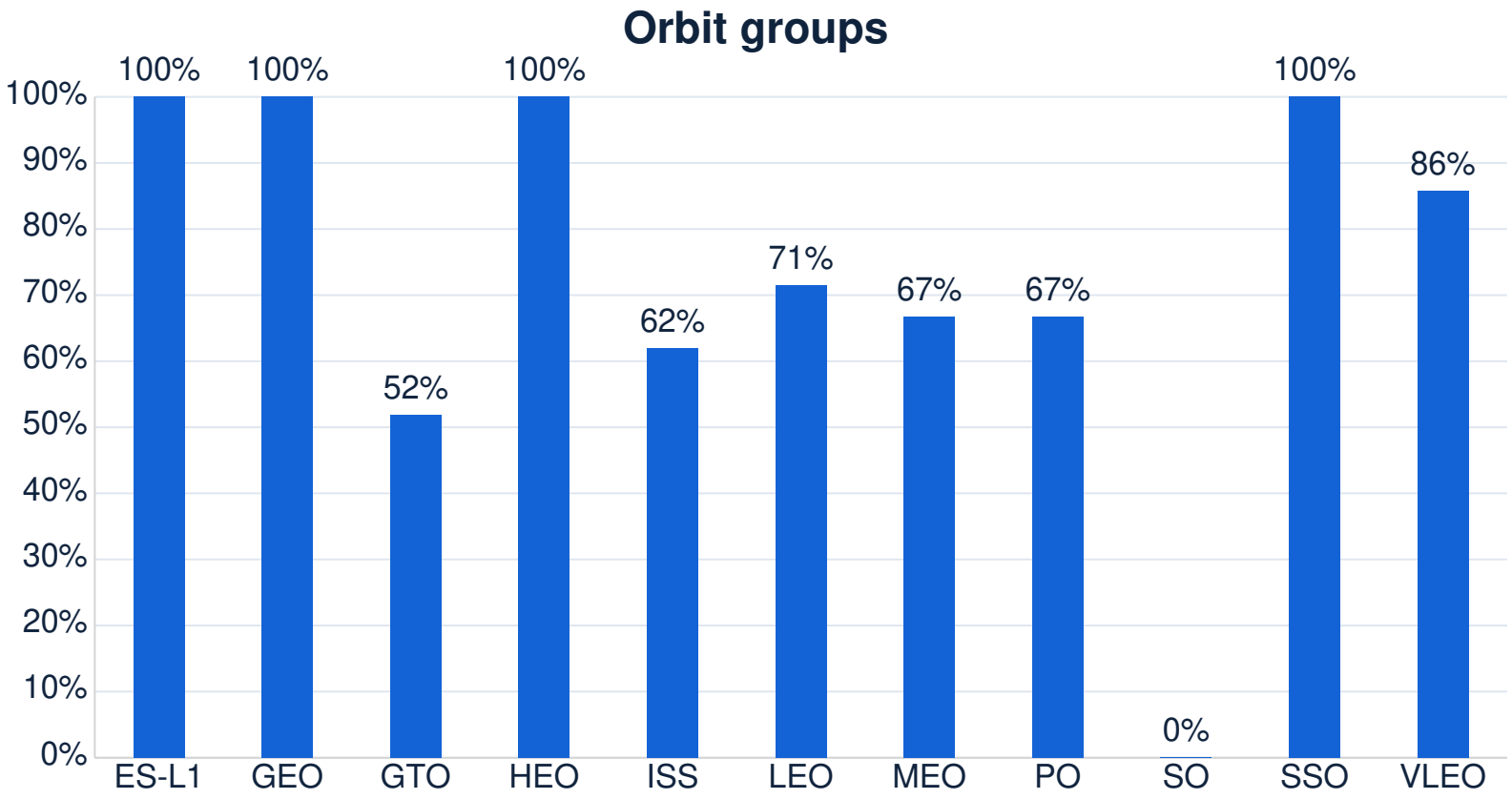
2019–2020
25 successful / 29 launches

The 2018 dip shows that progress is not strictly linear.

Early launches include missions without a landing attempt. Annual rates reflect both recovery capability and mission objectives.

IBM course datasets. Calculations reproduced for this report.

Landing Rates by Orbit



Several orbit groups have very few observations.

A 100% rate from one launch provides weak evidence.

Orbit relates to mission energy requirements, but this chart cannot isolate that effect.

Group sample sizes: ES-L1 1, GEO 1, GTO 27, HEO 1, ISS 21, LEO 7, MEO 3, PO 9, SO 1, SSO 5, VLEO 14

IBM course datasets. Calculations reproduced for this report.

SQL Results: Payloads and First Recovery

Query	Computed result
Total NASA (CRS) payload	45,596 kg
Mean payload, F9 v1.1 variants	2,534.67 kg
Maximum payload	15,600 kg across 12 records
First successful ground-pad landing	22 December 2015
Launch-site labels	4 distinct labels

Example: SELECT SUM(PAYLOAD_MASS__KG_) FROM launches WHERE Customer = 'NASA (CRS)';

IBM course datasets. Calculations reproduced for this report.

SQL Results: Outcomes and Failure Events

Finding	Computed result
Mission outcomes after trimming whitespace	99 success, 1 in-flight failure, 1 unclear payload status
Drone-ship failures during 2015	10 January and 14 April, both CCAFS LC-40
Successful drone recovery, 4–6 tonnes	4 booster records match the query
2010-06-04 to 2017-03-20 landing outcomes	10 no attempt, 5 drone success, 5 drone failure

Mission success and first-stage landing success are different labels. A mission can deliver its payload while the booster fails to land.

IBM course datasets. Calculations reproduced for this report.

Interactive Visual Analytics – Methodology

Folium: add launch-site circles, tooltips and a MarkerCluster layer. Color launch records green for success and red for class 0.

Use geographic coordinates to compare Cape Canaveral and Kennedy Space Center. A straight-line segment and haversine distance provide proximity analysis.

Dash: a launch-site dropdown and payload RangeSlider filter the same dataset. Callbacks update a success pie chart and payload/outcome scatter plot.

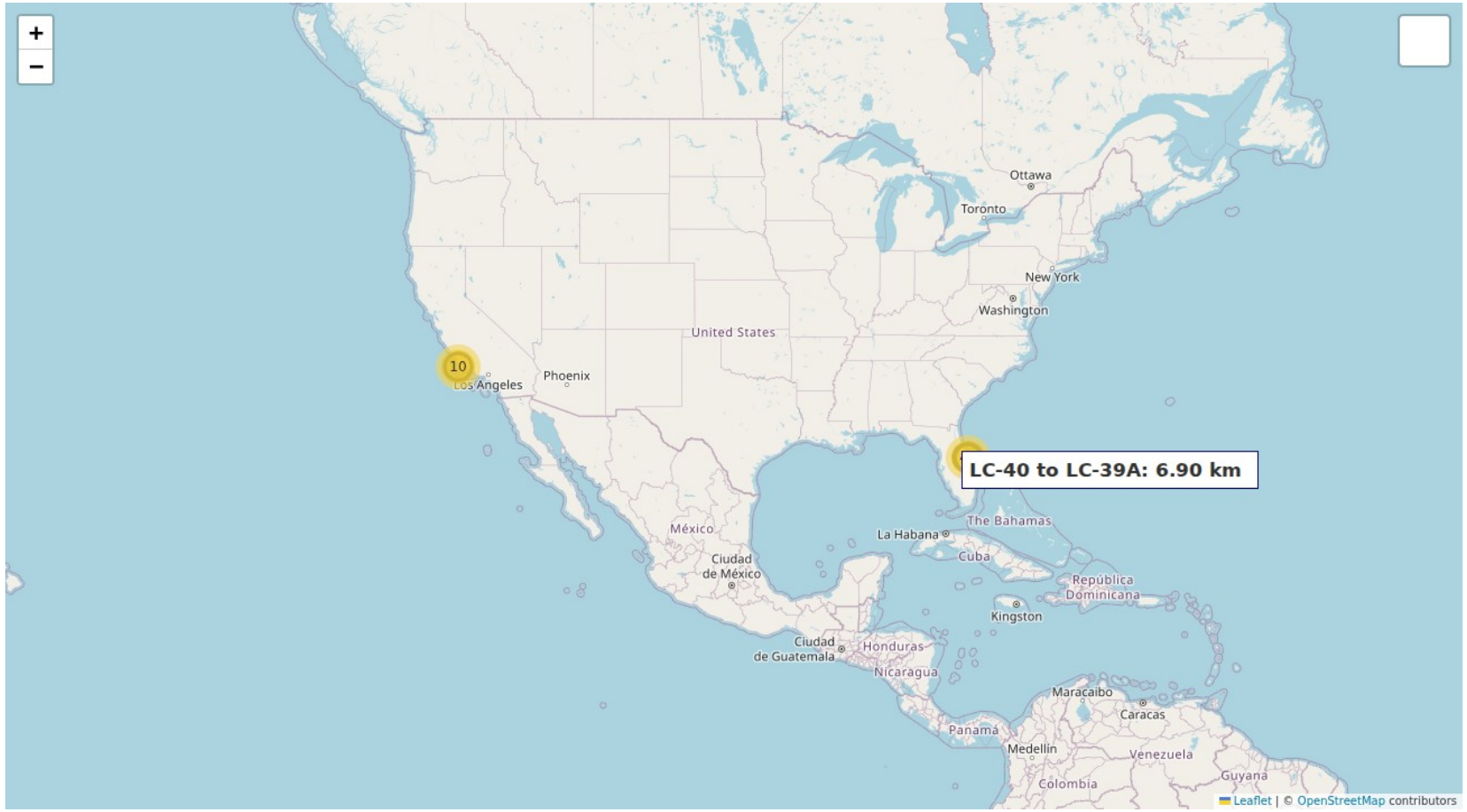
The interactive subset contains 56 launches and 24 successful landing outcomes. It differs from the 90-launch modeling snapshot.

GitHub: Folium notebook

GitHub: Plotly Dash application

IBM course datasets. Calculations reproduced for this report.

Folium Results: Launch Sites and Landing Records



Launch clusters appear on both US coasts.

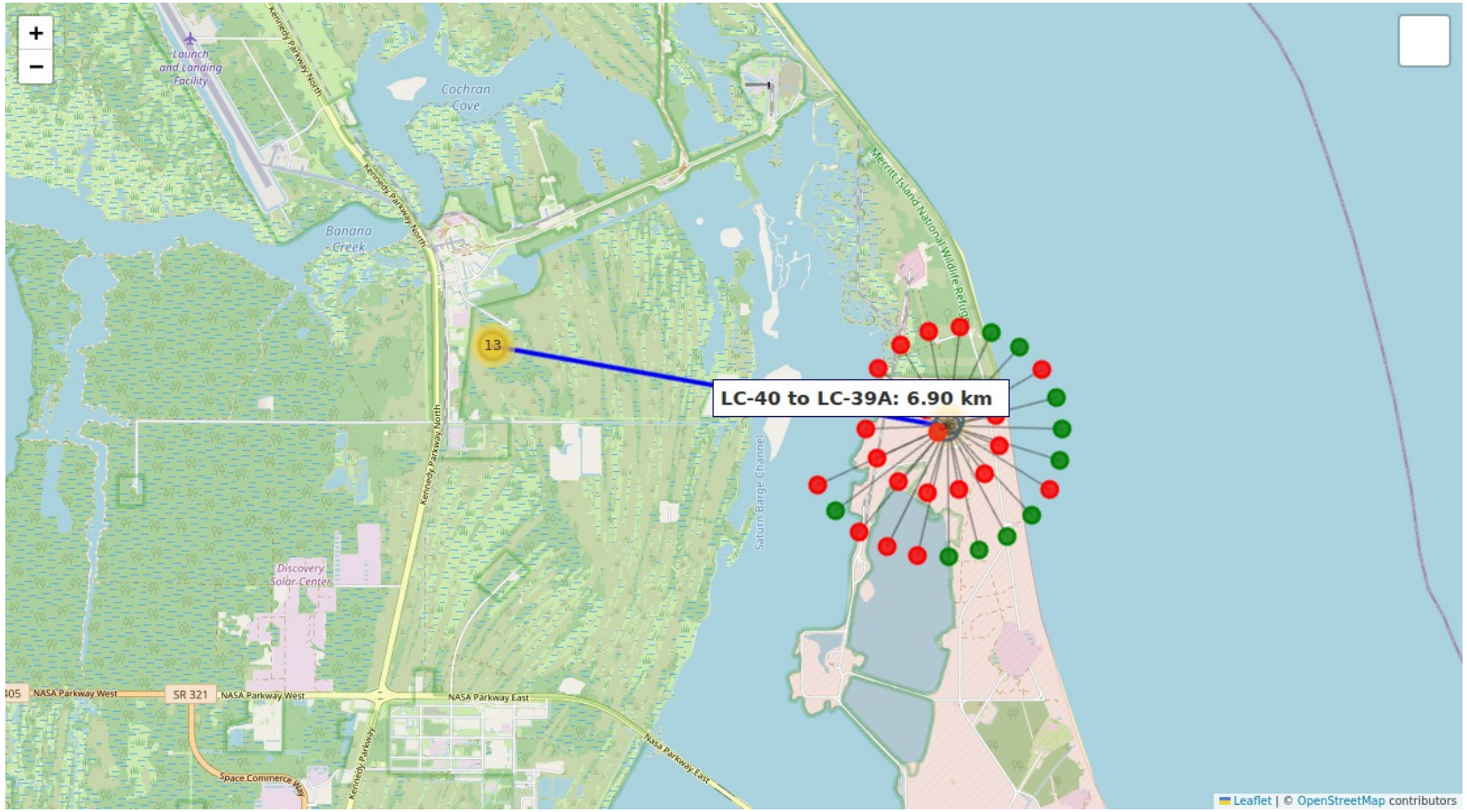
Green = successful landing. Red = unsuccessful, no attempt or unknown.

Marker clustering keeps overlapping launch records readable.

GitHub: Folium code and outputs

Actual Folium output. Source: IBM spacex_launch_geo.csv, 56 historical launches.

Folium Results: Proximity Analysis



The blue line connects Cape Canaveral LC-40 to KSC LC-39A.

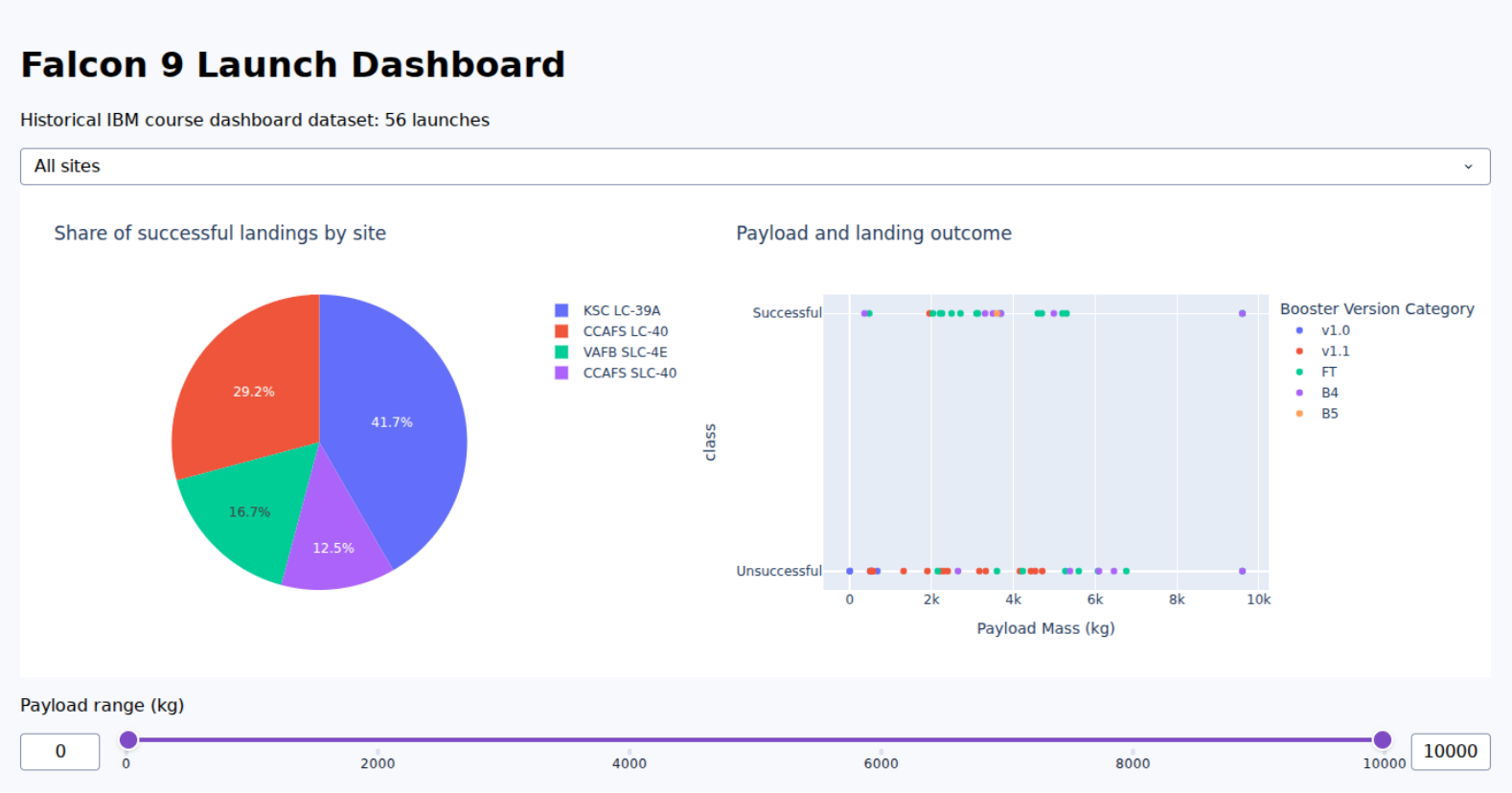
The map label shows the haversine distance between these nearby launch locations.

Colored records spread out to reveal overlap. The line gives a straight-line distance of 6.90 km.

GitHub: Interactive map generator

Straight-line geographic distance calculated from the course dataset coordinates.

Plotly Dash Results: All Sites



KSC contributes 10 of 24 successful landings: 41.7%.

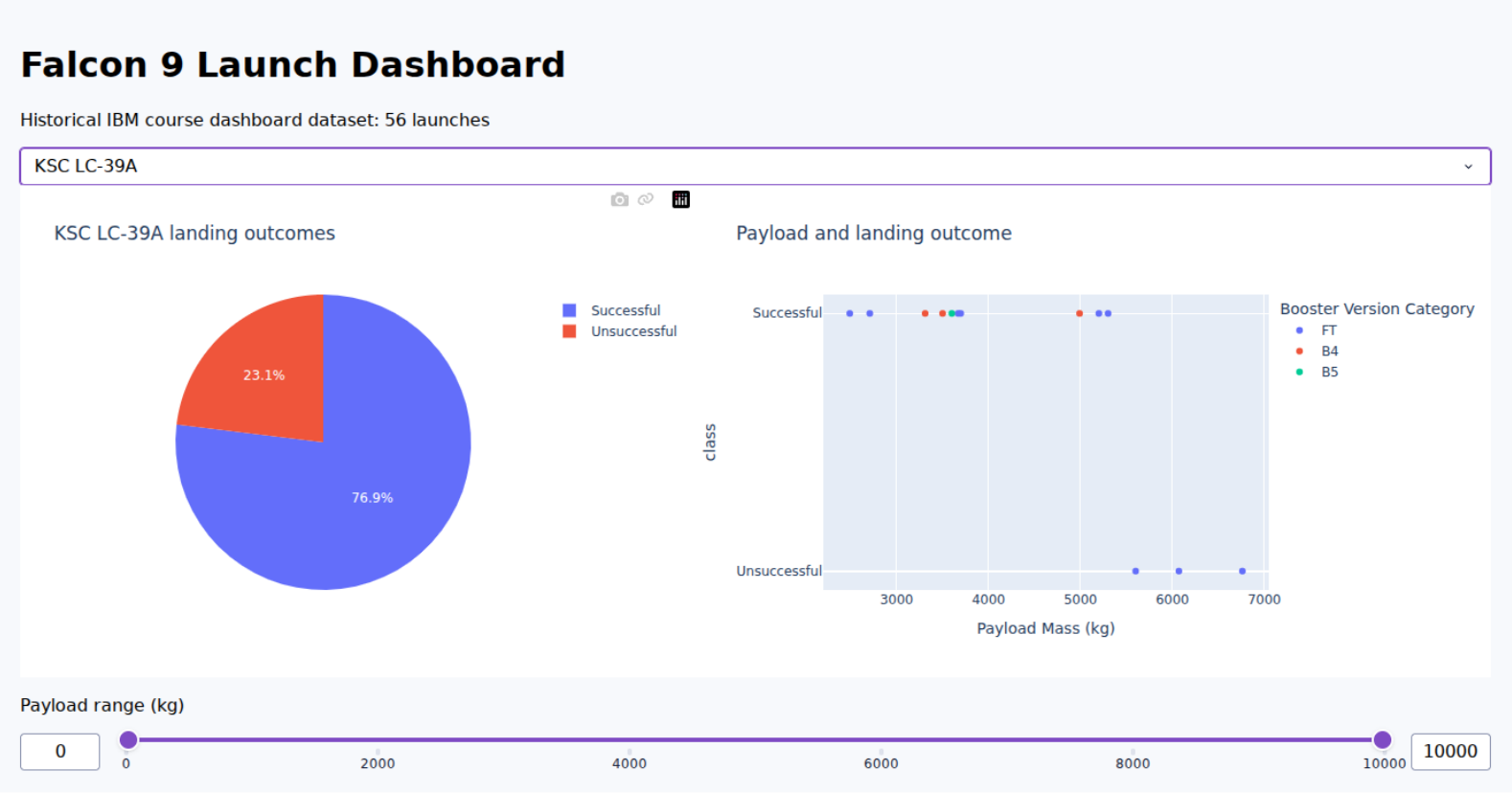
The scatter plot links payload, outcome and booster category.

Dropdown and range filters update both views.

GitHub: dashboard.py

Actual screenshot from the running Plotly Dash app. Source: 56-launch course dataset.

Plotly Dash Results: KSC and Payload Filtering



KSC LC-39A has 10 successes in 13 launches: 76.9%.

The site pie uses all selected launches as its denominator.

This differs from KSC's 41.7% share of all successful launches.

GitHub: dashboard.py

Actual selected-site dashboard screenshot. Payload filter applies to both charts.

Predictive Analysis Methodology

Split: 72 training launches and 18 held-out launches. Use stratified sampling with random_state = 2. The test set contains 12 positive and 6 negative labels.

Tune Logistic Regression, SVM, Decision Tree and KNN using five shuffled, stratified cross-validation folds on training data only.

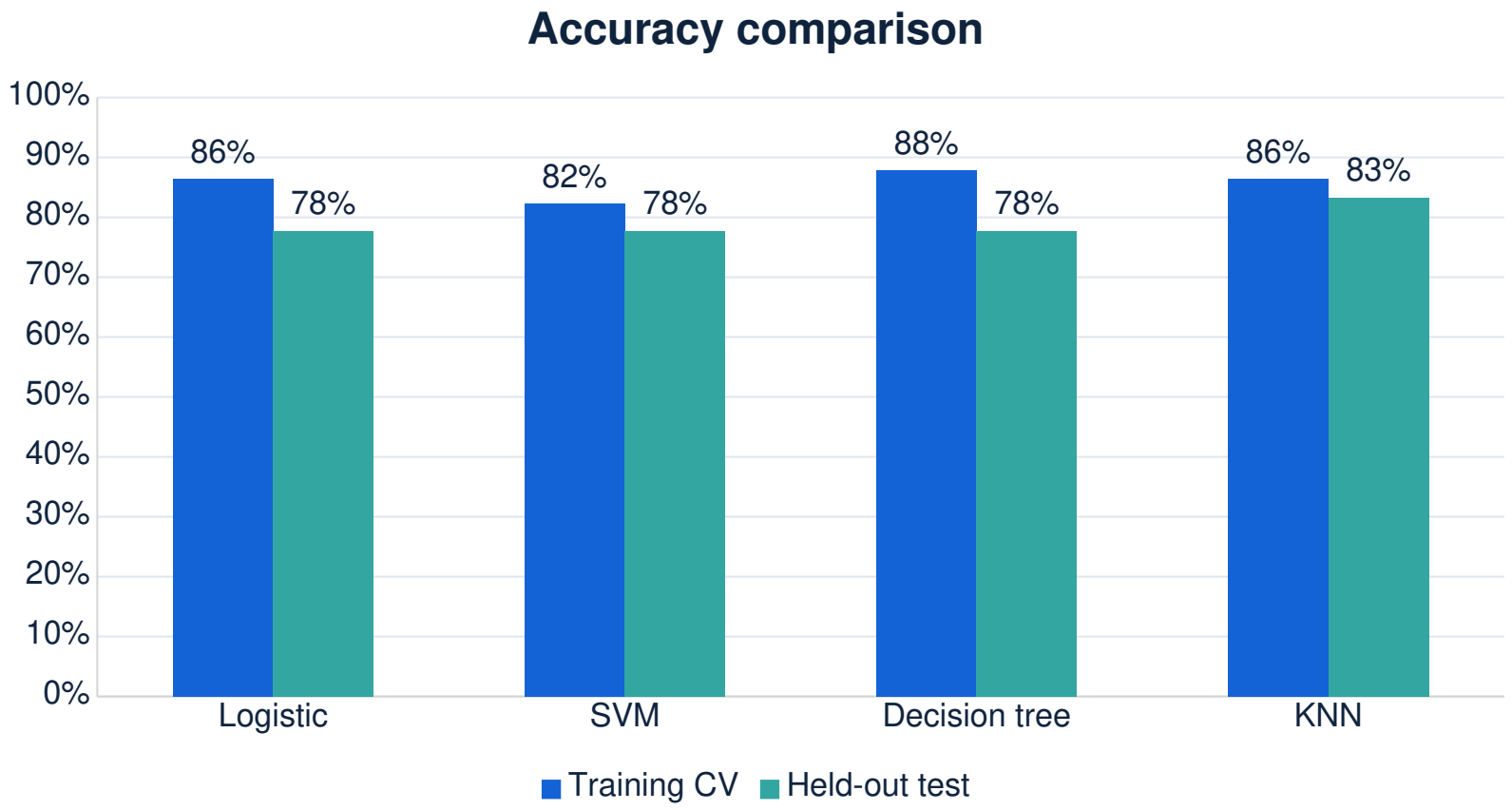
Pipeline: fit StandardScaler within each fold, then fit the classifier. Select the model using mean training cross-validation accuracy.

Compare held-out accuracy with a 66.7% always-positive baseline. Report confusion counts and positive-class precision, recall and F1.

GitHub notebook: 07_predictive_analysis.ipynb

IBM course datasets. Calculations reproduced for this report.

Classification Accuracy by Model



Decision tree: highest training CV, 87.8%.

KNN: highest test score, 83.3%.

Select using training CV to preserve the test as an evaluation set.

GitHub: model code, parameters and outputs

IBM course datasets. Calculations reproduced for this report.

Decision Tree Confusion Matrix

Actual / predicted	Class 0	Class 1
Class 0: unsuccessful	5	1
Class 1: successful	3	9

Accuracy: 14 / 18 = 77.8%. Positive precision: 90.0%. Recall: 75.0%. F1: 0.818.

One false positive overestimates landing success. Three false negatives underestimate it. A single test observation changes accuracy by 5.6 percentage points.

IBM course datasets. Calculations reproduced for this report.

Additional Insights and Validation Risks

Recovery target: True Ocean counts as successful in the course label. A commercially recovered booster target would need a different label definition.

Time matters: a separate first-72 / last-18 split gives 83.3% accuracy for the selected model family after tuning on the earlier training period.

Feature availability: booster serial identifiers and ReusedCount may encode historical information. A production model must verify what was known before each launch.

Next experiment: compare a prelaunch-only feature set with time-based and booster-grouped validation. Report uncertainty rather than a single accuracy estimate.

IBM course datasets. Calculations reproduced for this report.

Conclusion

Historical landing rates improve strongly between the early mission era and 2019–2020. Site and orbit comparisons provide useful descriptive context.

SQL confirms payload totals and important recovery milestones. Folium and Dash make the launch subsets easier to explore.

The selected decision tree exceeds the 66.7% baseline with 77.8% held-out accuracy, but 18 test launches provide limited precision.

A reliable operational model needs clearer recovery labels, a feature-availability audit and validation across future launches and unseen boosters.

IBM course datasets. Calculations reproduced for this report.

Repository and Reproducibility

GitHub repository: https://github.com/shyrradev/PYTHON/tree/ibm-spacex-capstone/ibm-spacex-capstone

The repository contains seven executed notebooks, Python analysis, all ten SQL queries, the Dash application, Folium map and data snapshots.

Reproduction: install requirements, run analysis.py, then launch dashboard.py. The results.json file records the computed tables and metrics.

The project occupies its own folder on the ibm-spacex-capstone branch of the public PYTHON repository.

https://github.com/shyrradev/PYTHON/tree/ibm-spacex-capstone/ibm-spacex-capstone

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